# Data Analytics Capstone

# Technical Declines in India's Unified Payments Interface: Effects of Volume Shifts and Regulatory Interventions

## Interim Report

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## GitHub Repository

The data files, code, and outputs for this report are in the public repository: https://github.com/karthick2613-git/ms_upi_capstone.

The repository holds the 33 original NPCI files (data/raw), the processed panel and audit files (data/processed), the scripts (code), the figures produced by the analysis (outputs), and the synopsis (docs). A README explains how to rerun every step. Paths are listed in the GitHub Data Availability Statement.

# Technical Declines in India's Unified Payments Interface: Effects of Volume Shifts and Regulatory Interventions

## Introduction

### Background and Context

The Unified Payments Interface (UPI) is the real-time retail payment system operated by the National Payments Corporation of India (NPCI). It lets customers of different banks move money instantly through third-party apps, and it has become the dominant way Indians pay for everyday goods and services. Academic work on UPI has mostly explained its growth (Balakrishnan, 2023; Gopika et al., 2026). Growth has been steep: in the NPCI member tables used here (NPCI, n.d.), the attempted volume of the bank members in the top 50 rose from 12.5 billion transactions in January 2024 to 26.5 billion in August 2026, so the system now carries about 2.1 times the load it did at the start of the study window.

Every UPI transaction that fails is classified by NPCI as a business decline (BD) or a technical decline (TD). A BD reflects a customer-side cause such as a wrong PIN or insufficient balance. A TD reflects a failure in the system itself, such as a connectivity, switch, or bank-side processing problem. NPCI reports the two separately for each member bank, together with the approved share, so that approved, BD, and TD percentages sum to 100.

Two forces make TD a live operational question. The first is load: when volume rises abruptly, a bank's processing capacity may be strained. The second is regulation. On April 12, 2025, a UPI outage was attributed by NPCI to an unexpected surge in Check Transaction Status application programming interface (API) calls (Tyagi, 2025). NPCI then announced measures on May 21, 2025, effective August 1, 2025, that cap balance checks at 50 per user per day per app, limit linked-account views to 25 per customer per day, allow at most three transaction-status checks in two hours with a minimum 90-second gap, and confine AutoPay mandate execution to non-peak hours (Khanduri & Mittal, 2025). The stakeholders are NPCI, which sets the rules, and remitter banks, which provision capacity. Whether abrupt growth and these controls moved technical declines is not documented in public bank-level evidence.

**Problem Statement**

Capacity decisions and API rules for UPI are made without bank-level evidence on what drives technical declines. Chawla et al. (2021) modeled how banking regulation could affect online payment failures, but, to our knowledge, no published study has used NPCI's member-level monthly performance tables to estimate how sudden volume shifts affect technical decline rates, or has evaluated the August 2025 API controls against a pre-intervention baseline. This research aims to estimate how sudden volume shifts and regulatory interventions affect technical decline rates in UPI, in order to help NPCI and banks decide where to add capacity and which API controls to enforce.

**Purpose of the Study**

The study is primarily explanatory and descriptive: it estimates the size and direction of associations between TD and volume shocks, calendar peaks, and the timing of the 2025 intervention, without claiming proof of cause, and it characterizes how persistent TD spikes are and how much of TD variation is specific to a bank. A secondary, predictive objective benchmarks machine learning (ML) against transparent baselines for next-month TD and spike detection. The dependent variable is the technical decline share, $TD_{it}$, the percentage of bank $i$'s attempted transactions in month $t$ that fail for technical reasons. The independent variables are sudden volume shifts (month-over-month log volume growth, $\Delta \ln V_{it}$, and an indicator $NewPeak_{it}$ that volume exceeds the bank's previous maximum), the intervention indicator $Post_t$ (1 from August 2025), and the calendar-peak indicator $Peak_t$ (March, October, and November).

**Interim Project Status (Progress Snapshot)**

- **Completed:** collection of 33 monthly NPCI files (32 months, January 2024–August 2026); cleaning, validation, and a documented panel of 1,598 rows; sample-size calculation for all four research questions; exploratory data analysis (EDA) with eight figures; five-specification interrupted time series with placebo breaks for RQ2; preliminary estimates for RQ1, RQ3, and RQ4; a rolling-origin benchmark of two naive baselines, a fractional logit, and XGBoost; public GitHub repository.

- **In progress:** robustness checks (leave-one-bank-out, alternative spike thresholds, winsorized outcome), wild cluster bootstrap inference for RQ1 and RQ3, and verification of two judgment calls in the data (a bank rename and the label of the August 2026 file).
- **Pending:** beneficiary-side tables and September 2026 data, hyperparameter tuning and final validation of the ML benchmark, a difference-in-differences contrast for RQ2 if beneficiary data allow, final recommendations, and the final report.

**Scope and Objectives**

The analysis covers the remitter side of UPI for the bank members in NPCI's top-50 list, monthly, from January 2024 to August 2026. The unit of analysis is a bank-month. In scope are RQ1–RQ4 below, the RQ2 timing evaluation with placebo breaks, and the ML benchmark. Out of scope are transaction-level or daily data (not public), user-level behavior, merchant-side analysis, and any claim that the August 2025 controls are proven to cause a change in TD. The scope is feasible because the data already exist in compiled form and the models are standard open-source procedures. The research questions and their measurable objectives are as follows.

***Research Question 1 (RQ1)***

Do sudden volume shifts raise a bank's technical decline rate? $H0_{1a}$: the coefficient on $\Delta \ln V_{it}$ is zero; $H1_{1a}$: it is not zero. $H0_{1b}$: the NewPeak coefficient is zero; $H1_{1b}$: it is not. The objective is to estimate the marginal effect, in percentage points (pp), of a 10% volume jump and of a new peak on TD, with 95% confidence intervals (CI).

***Research Question 2 (RQ2)***

Did system-level TD change after NPCI's API controls took effect on August 1, 2025? $H0_2$: the post-intervention level shift in volume-weighted monthly TD is zero; $H1_2$: it is not zero. Because every bank was exposed on the same date, this is a timing evaluation with a pre-set rule: an August-specific association is reported only if sign and 95% interval agree across five specifications and August ranks in the top two of 21 candidate break months.

***Research Question 3 (RQ3)***

Do predictable calendar peaks (March fiscal year-ends and the October–November festive months) raise TD, and do high-volume banks suffer more? $H0_{3a}$: TD in peak months equals TD in other months for the same bank; $H0_{3b}$: the peak effect does not vary with a bank's volume exposure.

***Research Question 4 (RQ4)***

How persistent are TD spikes, and how much of TD variation is specific to the bank? $H0_{4a}$: the bank-level share of TD variance (the intraclass correlation, ICC) is zero; $H0_{4b}$: month-to-month autocorrelation of TD within a bank is zero; $H0_{4c}$: spike recovery times do not differ across bank groups.

## Literature Survey

### Literature Review Approach

Sources were found through Google Scholar, RePEc/IDEAS, the MPRA archive, and publisher and regulator sites, using keywords such as UPI, technical decline, payment failure, outage, interrupted time series, fractional response, and survival. Sources were included if they were peer-reviewed, a regulator publication, an archived working paper, or clearly attributed commentary on a UPI event, and bore on the UPI setting, payment reliability, or a method used here. Fourteen sources are summarized in Table 1, and the full reference list is at the end of the report. Searches for studies estimating the effect of volume shifts and the 2025 controls on bank-level technical declines returned none, so the gap is worded as "to our knowledge."

**Table 1**
*Literature Relevance Matrix*

| Author (Year) | Domain / context | Dataset / setting | Method(s) | Key findings | Support for RQs and decisions |
|---|---|---|---|---|---|
| Balakrishnan (2023) | UPI growth | India, digital payments since 2010; Brazil's PIX for comparison | Descriptive and comparative analysis | UPI is the main driver of India's payment growth; the paper also discusses UPI's challenges | Context for volume as exposure; frames challenges of scale |
| Gopika et al. (2026) | UPI growth drivers | Monthly NPCI UPI volume, July 2016–May 2022 | ARDL error-correction model | Internet penetration, app count, and a COVID-19 structural break raise volume; POS terminals and cash relate negatively | Treats volume as the outcome; this study treats it as exposure. Shows NPCI monthly data and structural breaks are usable (RQ2) |
| Chawla et al. (2021) | Banking regulation and payment failure | Simulated bank cost model; no bank-level data | Theoretical profit model with simulations | Bank profit depends on how held funds are invested; stricter penalties can produce losses | Closest topic, but theoretical; leaves the empirical question of RQ1 and RQ2 open |
| Griffiths & Joyce (2024) | Payment reliability, Australia | RBA incident data, September 2021–March 2024 | Descriptive monitoring of availability and outages | Services averaged at least 99.8% availability per quarter; third-party issues were the leading cause; a few institutions account for many incidents | Supports a bank-heterogeneity lens (RQ4) and the use of availability metrics |
| Tyagi (2025) | UPI outage, policy commentary | April 2025 outage in India | Qualitative policy analysis | Ties the outage to a surge in API calls and argues for resilience | Motivates RQ2; defines the intervention context |
| Khanduri & Mittal (2025) | UPI API rules, legal commentary | NPCI guidelines effective August 1, 2025 | Regulatory summary | Lists caps on balance checks, status checks, and AutoPay timing | Defines the intervention date and the control list for RQ2 |
| Bernal et al. (2017) | Policy evaluation methods | Public-health intervention examples | Interrupted time series tutorial | Level and slope changes can be estimated from a series; autocorrelation, seasonality, and control series matter | Basis for the RQ2 design |
| Wagner et al. (2002) | Policy evaluation methods | Medication-use policy studies | Segmented regression | Describes level and trend change with autocorrelation adjustment | Supports the RQ2 level-plus-trend specification |
| Abadie et al. (2010) | Comparative case studies | Synthetic control application to a state tobacco program | Synthetic control with placebo tests | Placebo analyses give a reference distribution for how unusual an estimate is | Motivates the placebo-in-time test (RQ2) |

| Author (Year) | Domain / context | Dataset / setting | Method(s) | Key findings | Support for RQs and decisions |
|---|---|---|---|---|---|
| Papke & Wooldridge (1996) | Econometrics | 401(k) plan participation rates (proportions) | Quasi-likelihood fractional logit | Bounded proportions can be modeled directly without impossible fitted values | Model for RQ1 and RQ3 |
| Cameron et al. (2008) | Econometrics | Simulations and applications with few clusters | Wild cluster bootstrap | Clustered tests over-reject with few clusters; bootstrap corrections help | Inference plan for about 56 bank clusters |
| Kaplan & Meier (1958); Cox (1972) | Survival analysis | Incomplete observation of durations | Product-limit estimator; proportional hazards | Durations and covariate effects can be estimated with censoring | RQ4 spike duration |
| Chen & Guestrin (2016) | Machine learning | Large-scale benchmarks | XGBoost gradient-boosted trees | A scalable tree-boosting method with strong benchmark results | ML benchmark model |
| Bergmeir & Benítez (2012); Saito & Rehmsmeier (2015) | Evaluation methods | Time series predictors; imbalanced classification | Order-respecting validation; precision–recall analysis | Time-ordered validation suits time series; precision–recall is more informative than ROC when positives are rare | Validation and metric choices for the benchmark |

*Note.* ARDL = autoregressive distributed lag; RBA = Reserve Bank of Australia; ROC = receiver operating characteristic.

The literature falls into four themes. The first is growth and adoption. Balakrishnan (2023) traces India's digital-payments expansion and compares UPI with Brazil's PIX, and Gopika et al. (2026) estimate the drivers of UPI volume with an error-correction model on monthly NPCI data. Both treat volume as the outcome and neither examines service quality. Their value for this study is twofold: they confirm that NPCI's monthly tables support time-series modeling, and they show that structural breaks matter, since Gopika et al. find a COVID-19 break.

The second theme is reliability and failure. Griffiths and Joyce (2024) report availability by service from regulator-collected incident data, find that third-party causes dominate, and note that a few institutions account for a large share of incidents. That finding supports looking at bank heterogeneity and spike persistence (RQ4). Chawla et al. (2021) is the closest topic to this project, but it is a theoretical profit model with simulated inputs, not an analysis of observed

bank data. Tyagi (2025) and Khanduri and Mittal (2025) describe the April 2025 outage and the controls that followed. They establish the events and dates but estimate no effect on technical declines.

The third theme is quasi-experimental and panel methodology. Interrupted time series is the standard design for a policy introduced at a known date (Bernal et al., 2017; Wagner et al., 2002), but with a single system and one date it cannot rule out coincident shocks, which is why this project adds placebo breaks in the spirit of Abadie et al. (2010) and a pre-set decision rule. Fractional logit suits TD because it is a proportion with many values near zero (Papke & Wooldridge, 1996), and with about 56 banks, cluster-robust tests need bootstrap corrections (Cameron et al., 2008). Survival methods suit spike durations (Cox, 1972; Kaplan & Meier, 1958).

The fourth theme is prediction and evaluation. Gradient-boosted trees (Chen & Guestrin, 2016) capture nonlinearities but give no direct effect estimates, and with rare events precision–recall measures are more informative than accuracy or ROC curves (Saito & Rehmsmeier, 2015). Because the data are ordered in time, validation must respect order (Bergmeir & Benítez, 2012). The gap is clear: the UPI literature explains volume but not failure, the failure literature is non-Indian or theoretical, and the methods have not been applied to NPCI's bank-level tables. This project pairs transparent models for RQ1–RQ4 with a boosted-tree benchmark that tests whether they miss structure.

## Data Description

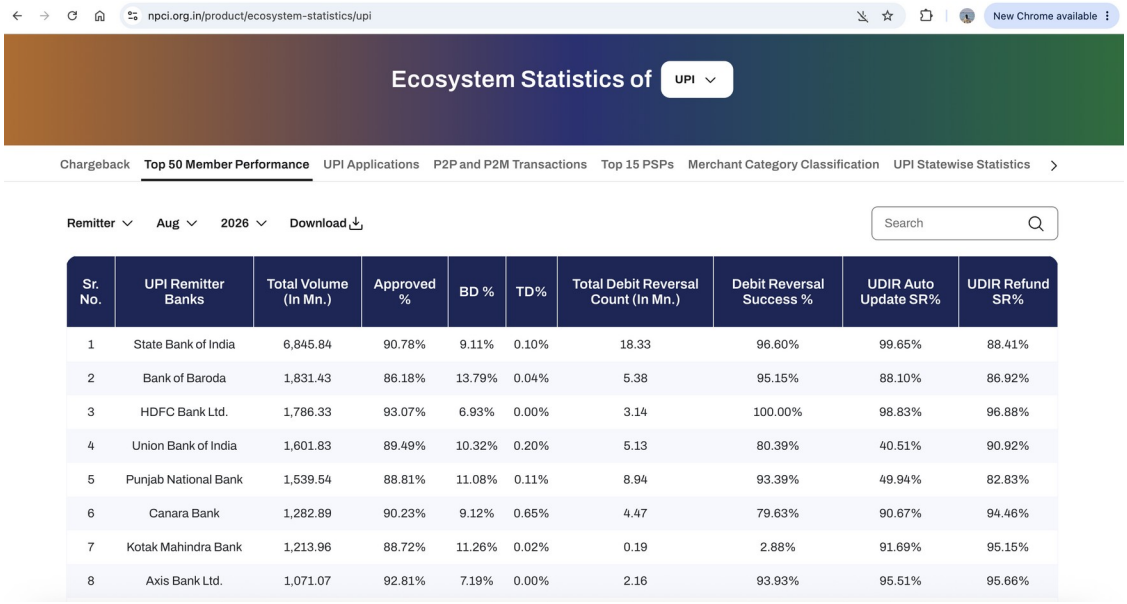
### Data Source(s) and Access

The data are NPCI's public "Top 50 Member Performance" tables for the remitter side, available at https://www.npci.org.in/product/ecosystem-statistics/upi (NPCI, n.d.). They are real, public, aggregate statistics, not synthetic data. NPCI publishes one table per month; the portal offers downloads from January 2024, and the 32 months used run from January 2024 to August 2026. Because NPCI does not permit automated access, the files were downloaded manually and

combined with a documented script. Figure 1 shows the portal page, and the original files are in the repository.

**Figure 1**

*NPCI UPI Ecosystem Statistics Page, Top 50 Member Performance (Remitter, August 2026)*




| Sr. No. | UPI Remitter Banks | Total Volume (In Mn.) | Approved % | BD % | TD% | Total Debit Reversal Count (In Mn.) | Debit Reversal Success % | UDIR Auto Update SR% | UDIR Refund SR% |
|---|---|---|---|---|---|---|---|---|---|
| 1 | State Bank of India | 6,845.84 | 90.78% | 9.11% | 0.10% | 18.33 | 96.60% | 99.65% | 88.41% |
| 2 | Bank of Baroda | 1,831.43 | 86.18% | 13.79% | 0.04% | 5.38 | 95.15% | 88.10% | 86.92% |
| 3 | HDFC Bank Ltd. | 1,786.33 | 93.07% | 6.93% | 0.00% | 3.14 | 100.00% | 98.83% | 96.88% |
| 4 | Union Bank of India | 1,601.83 | 89.49% | 10.32% | 0.20% | 5.13 | 80.39% | 40.51% | 90.92% |
| 5 | Punjab National Bank | 1,539.54 | 88.81% | 11.08% | 0.11% | 8.94 | 93.39% | 49.94% | 82.83% |
| 6 | Canara Bank | 1,282.89 | 90.23% | 9.12% | 0.65% | 4.47 | 79.63% | 90.67% | 94.46% |
| 7 | Kotak Mahindra Bank | 1,213.96 | 88.72% | 11.26% | 0.02% | 0.19 | 2.88% | 91.69% | 95.15% |
| 8 | Axis Bank Ltd. | 1,071.07 | 92.81% | 7.19% | 0.00% | 2.16 | 93.93% | 95.51% | 95.66% |

*Note.* Screenshot taken by the author from NPCI (n.d.).

## Dataset Overview

Each row is one member in one month, with attempted volume, the approved, BD, and TD shares, and debit-reversal fields. The combined panel is summarized below.

- Number of records (rows): 1,598 (1,555 bank rows and 43 rows for credit-card product lines and non-bank payment firms, which are flagged and excluded from bank-level models).
- Number of variables (columns): 18 after cleaning (11 NPCI-derived fields, 6 identifiers and audit fields, and the raw-name field), plus 6 derived analysis variables.
- Time period: January 2024 to August 2026, 32 consecutive months; 30 months list 50 members and 2 list 49 because two duplicate listings were combined.
- Unit of analysis: bank-month; 62 entities after one rename mapping, of which 56 are banks.

- Target variable: TD share, the percentage of attempted transactions declined for technical reasons (td_pct).

**Data Dictionary**

Table 2 defines every field, its type, allowed values, the handling of missing values, and notes. The last six rows are derived from the NPCI fields.

**Table 2**
*Data Dictionary*

| Variable | Definition | Datatype | Allowed values / range | Missing values handling | Notes |
|---|---|---|---|---|---|
| month | Reporting month | Date (YYYY-MM) | 2024-01 to 2026-08 | None missing | Taken from the sheet title; one file by filename |
| rank | NPCI rank by volume in the month | Integer | 1–50 | None missing | Descriptive |
| bank_raw, bank | Name as published; harmonized name | Text | 159 raw spellings; 63 harmonized names | None missing | Rule-based map |
| entity | Bank after merger or rename mapping | Text | 62 entities (56 banks) | None missing | Cluster and fixed-effect unit |
| volume_mn | Attempted remitter transactions | Numeric | Observed 14.1–6,846 million | None missing | Exposure and weight |
| approved_pct | Share of attempts approved | Numeric | 0–100 (observed 54.1–97.2) | None missing | Check: sums with BD and TD to 100 |
| bd_pct | Business decline share | Numeric | 0–100 (observed 2.2–27.0) | None missing | Placebo outcome |
| td_pct | Technical decline share | Numeric | 0–100 (observed 0.0–36.3) | None missing; 70 zeros kept, floored at 0.0025 pp for logits | Dependent variable |
| debit_reversal_mn | Debit reversals | Numeric | 0–23.2 million (observed) | 29 missing (1.8%); left blank | Not used in models |
| debit_reversal_success_pct | Reversals completed successfully | Numeric | 0–100 | 3 missing (0.2%); left blank | Not used in models |
| is_bank | False for card product lines and non-bank firms | Logical | True / False | None missing | Filter (43 rows False) |
| source_file, month_source | File used and how month was set | Text | 33 files | None missing | Audit trail |

| Variable | Definition | Datatype | Allowed values / range | Missing values handling | Notes |
|---|---|---|---|---|---|
| lnV, ΔlnV | Log volume; month-over-month log growth | Numeric | ΔlnV observed −1.10 to 0.81 | ΔlnV missing for each bank's first month (excluded in RQ1) | Independent (RQ1) |
| NewPeak | 1 if volume exceeds the bank's previous maximum | Binary | 0 / 1 | Undefined in first month | Independent (RQ1) |
| Post, Peak | Post: 1 from 2025-08. Peak: 1 in March, October, November | Binary | 0 / 1 | None missing | Independent (RQ2, RQ3) |
| Spike | 1 if robust z of TD exceeds 3 and TD is at least 1.0% | Binary | 0 / 1 | Bank with zero MAD: z undefined, no spike | Outcome (RQ4) |

*Note.* Approved + BD + TD = 100% in every row (maximum deviation 0.01 pp). MAD = median absolute deviation. Source: NPCI (n.d.).

## GitHub Data Availability Statement

The raw and processed datasets are available in the GitHub repository at https://github.com/karthick2613-git/ms_upi_capstone under the paths below. The repository is public, so an evaluator can open every file without access rights.

- Raw data: data/raw (33 original NPCI Excel files, unaltered).
- Processed/clean data: data/processed (upi_remitter_top50_panel.csv, file_log.csv, month_audit.csv, entity_crosswalk.csv).
- Code/notebooks: code (combine_remitter.py, sample_size.py, rq2_robustness.py, interim_analysis.py and result files); figures in outputs.

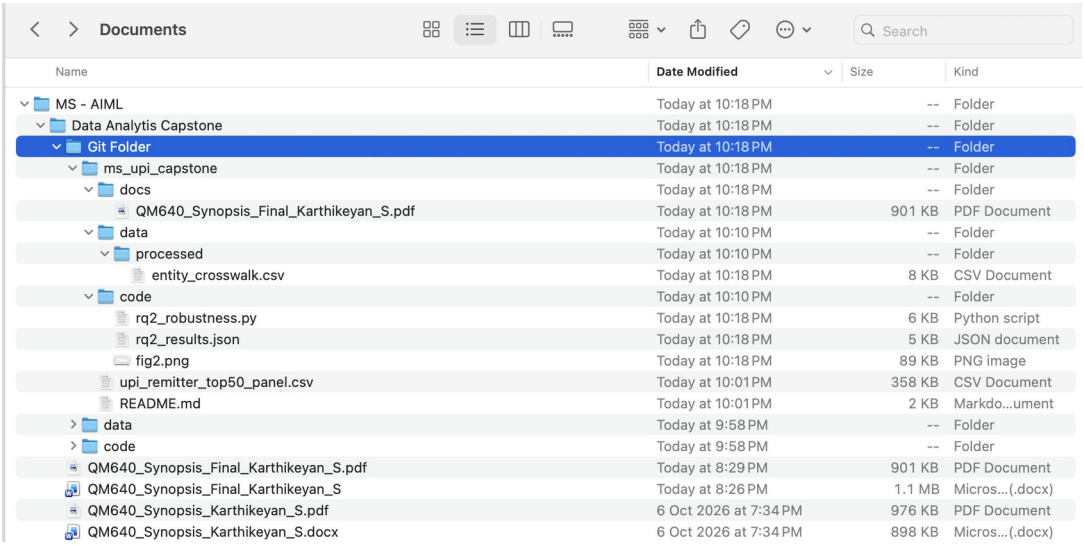
## Screenshots and Evidence

Figure 2 previews the first rows of the processed panel, and Figure 3 shows the repository folder structure on the author's computer after the final push.

**Figure 2**
*First Rows of the Combined Remitter Panel (upi_remitter_top50_panel.csv)*

| month | file_month | rank | bank_raw | bank | volume_mn | approved_pc | bd_pct | td_pct | debit_reversa | debit_reversa | source_file | month_sourc | n_member_r | entity | is_bank | pct_sum |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2024-01 | 2024-01 | 1 | State Bank O | State Bank of | 3128.97 | 95.62 | 4.19 | 0.19 | 11.58 | 96.85 | b637e410-Ec | sheet title | 1 | State Bank of | TRUE | 100 |
| 2024-01 | 2024-01 | 2 | HDFC BANK | HDFC Bank | 1040.25 | 96.49 | 3.51 | 0 | 3.34 | 98.84 | b637e410-Ec | sheet title | 1 | HDFC Bank | TRUE | 100 |
| 2024-01 | 2024-01 | 3 | Bank of Baro | Bank of Baro | 781.72 | 95.28 | 4.68 | 0.04 | 2.71 | 98.54 | b637e410-Ec | sheet title | 1 | Bank of Baro | TRUE | 100 |
| 2024-01 | 2024-01 | 4 | Union Bank o | Union Bank o | 753.66 | 94.51 | 4.34 | 1.15 | 7.16 | 46.76 | b637e410-Ec | sheet title | 1 | Union Bank o | TRUE | 100 |
| 2024-01 | 2024-01 | 5 | Punjab Natio | Punjab Natio | 618.95 | 94.78 | 4.95 | 0.27 | 2.61 | 88.58 | b637e410-Ec | sheet title | 1 | Punjab Natio | TRUE | 100 |
| 2024-01 | 2024-01 | 6 | Kotak Mahinc | Kotak Mahinc | 600.83 | 93.64 | 5.19 | 1.17 | 2.08 | 89.49 | b637e410-Ec | sheet title | 1 | Kotak Mahinc | TRUE | 100 |
| 2024-01 | 2024-01 | 7 | Canara Bank | Canara Bank | 572.16 | 94.18 | 5.38 | 0.44 | 3.02 | 85.98 | b637e410-Ec | sheet title | 1 | Canara Bank | TRUE | 100 |
| 2024-01 | 2024-01 | 8 | Axis Bank Ltd | Axis Bank | 559.49 | 96.11 | 3.58 | 0.3 | 2.66 | 60.21 | b637e410-Ec | sheet title | 1 | Axis Bank | TRUE | 99.99 |
| 2024-01 | 2024-01 | 9 | ICICI Bank | ICICI Bank | 547.55 | 96.59 | 3.32 | 0.08 | 1.64 | 95.72 | b637e410-Ec | sheet title | 1 | ICICI Bank | TRUE | 99.99 |
| 2024-01 | 2024-01 | 10 | Paytm Payme | Paytm Payme | 410.06 | 96.3 | 3.7 | 0.01 | 0.87 | 98.11 | b637e410-Ec | sheet title | 1 | Paytm Payme | TRUE | 100.01 |
| 2024-01 | 2024-01 | 11 | Bank of India | Bank of India | 389.29 | 95.33 | 4.59 | 0.08 | 1.48 | 91.49 | b637e410-Ec | sheet title | 1 | Bank of India | TRUE | 100 |
| 2024-01 | 2024-01 | 12 | Indian Bank | Indian Bank | 348.89 | 95.15 | 4.47 | 0.37 | 1.51 | 84.37 | b637e410-Ec | sheet title | 1 | Indian Bank | TRUE | 99.99 |
| 2024-01 | 2024-01 | 13 | Airtel Payme | Airtel Payme | 337.05 | 89.95 | 8.98 | 1.07 | 1.93 | 69.9 | b637e410-Ec | sheet title | 1 | Airtel Payme | TRUE | 100 |
| 2024-01 | 2024-01 | 14 | India Post Pa | India Post Pa | 265.13 | 82.66 | 12.2 | 5.14 | 1.67 | 54.81 | b637e410-Ec | sheet title | 1 | India Post Pa | TRUE | 100 |
| 2024-01 | 2024-01 | 15 | Federal Bank | Federal Bank | 185.9 | 95.53 | 4.28 | 0.19 | 0.65 | 93.72 | b637e410-Ec | sheet title | 1 | Federal Bank | TRUE | 100 |
| 2024-01 | 2024-01 | 16 | Central Bank | Central Bank | 178.41 | 85.44 | 9.38 | 5.18 | 4.29 | 72.15 | b637e410-Ec | sheet title | 1 | Central Bank | TRUE | 100 |
| 2024-01 | 2024-01 | 17 | Indian Overs | Indian Overs | 165.09 | 94.28 | 4.41 | 1.31 | 0.99 | 90.36 | b637e410-Ec | sheet title | 1 | Indian Overs | TRUE | 100 |
| 2024-01 | 2024-01 | 18 | UCO Bank | UCO Bank | 152.41 | 88.37 | 7.47 | 4.16 | 2.33 | 40.45 | b637e410-Ec | sheet title | 1 | UCO Bank | TRUE | 100 |
| 2024-01 | 2024-01 | 19 | IDBI Bank Lin | IDBI Bank | 141.57 | 92.73 | 4.28 | 2.98 | 1.31 | 81.4 | b637e410-Ec | sheet title | 1 | IDBI Bank | TRUE | 99.99 |
| 2024-01 | 2024-01 | 20 | Bank of Mahc | Bank of Maha | 136.29 | 94.42 | 4.61 | 0.97 | 0.61 | 76.33 | b637e410-Ec | sheet title | 1 | Bank of Maha | TRUE | 100 |
| 2024-01 | 2024-01 | 21 | INDUSIND B | IndusInd Ban | 125.63 | 93.06 | 5.97 | 0.97 | 0.68 | 98.66 | b637e410-Ec | sheet title | 1 | IndusInd Ban | TRUE | 100 |
| 2024-01 | 2024-01 | 22 | Fino Paymen | Fino Paymen | 115.14 | 89.66 | 9.41 | 0.93 | 0.63 | 67.81 | b637e410-Ec | sheet title | 1 | Fino Paymen | TRUE | 100 |
| 2024-01 | 2024-01 | 23 | Karnataka Ba | Karnataka Ba | 97.36 | 91.93 | 7.49 | 0.59 | 0.42 | 77.55 | b637e410-Ec | sheet title | 1 | Karnataka Ba | TRUE | 100.01 |
| 2024-01 | 2024-01 | 24 | Yes Bank Ltd | YES Bank | 93.47 | 94.92 | 4.98 | 0.1 | 0.25 | 93 | b637e410-Ec | sheet title | 1 | YES Bank | TRUE | 100 |
| 2024-01 | 2024-01 | 25 | IDFC FIRST B | IDFC FIRST B | 73.92 | 94.2 | 5.77 | 0.03 | 0.03 | 0 | b637e410-Ec | sheet title | 1 | IDFC FIRST B | TRUE | 100 |
| 2024-01 | 2024-01 | 26 | Karur Vysya B | Karur Vysya B | 59.56 | 94.12 | 5.54 | 0.34 | 0.24 | 92.86 | b637e410-Ec | sheet title | 1 | Karur Vysya B | TRUE | 100 |
| 2024-01 | 2024-01 | 27 | Bandhan Bar | Bandhan Bar | 49.22 | 87.51 | 6.16 | 6.32 | 0.85 | 68.2 | b637e410-Ec | sheet title | 1 | Bandhan Bar | TRUE | 99.99 |
| 2024-01 | 2024-01 | 28 | South Indian | South Indian | 48.9 | 96.35 | 3.55 | 0.11 | 0.14 | 95.97 | b637e410-Ec | sheet title | 1 | South Indian | TRUE | 100.01 |
| 2024-01 | 2024-01 | 29 | Jammu and K | Jammu & Kas | 44.46 | 91.55 | 7.21 | 1.24 | 0.31 | 70.87 | b637e410-Ec | sheet title | 1 | Jammu & Kas | TRUE | 100 |

**Figure 3**
*Repository Folder Structure*




*Note.* data/raw holds the 33 original files, data/processed the panel and audit files, code the scripts, and docs the synopsis.

## Completeness and Limitations of the Data

The panel is complete for the study window: no core variable (volume, approved, BD, or TD share) is missing, and every month has exactly one source file. Three limitations matter. First, the top-50 list changes, so the panel is unbalanced: only 39 of 56 bank entities appear in all 32 months. Second, the data are monthly aggregates by bank, so findings describe banks, not users or transactions, and a single monthly TD rate cannot show when within the month a failure

occurred. Third, NPCI defines TD, and a change in classification practice would look like a real change in failure rates; this cannot be ruled out from public data. Two judgment calls are flagged: merging Andhra Pradesh Grameena Vikas Bank into Telangana Grameena Bank (12 of 1,555 bank-months, 0.8%; see the entity crosswalk), and the month label of the August 2026 file, whose sheet title says July but whose content differs from the July file.

**Sample Size and Power**

Sample size was set for each question using power or confidence-interval logic. Calculations use $\alpha = 0.05$ (two-sided, $z = 1.960$) and power of 0.80 ($z = 0.842$), so $K = 1.960 + 0.842 = 2.802$. For RQ1, a slope power formula, $n = (K\sigma_e / (\beta\, SD_x))^2$, is used with $\sigma_e = 1.0508$ (the residual SD of the logit of TD in a model with bank and month effects), $SD_x = 0.1258$ (the within-bank SD of $\ln V$), and a smallest effect of interest $\beta = 0.60$ logit units per unit of $\ln V$. This value is a planning choice: a 10% volume increase would then raise the odds of a technical decline by $\exp(0.60 \times \ln 1.1) - 1 = 5.9\%$. It gives $n = (2.802 \times 1.0508 / (0.60 \times 0.1258))^2 = 1{,}521.8$, rounded up to 1,522. For RQ2, a confidence-interval criterion asks for a 95% half-width of at most 0.25 pp for the level change; with $\sigma = 0.1353$ pp and lag-1 autocorrelation $\rho = 0.2027$, the variance inflation is $(1 + \rho) / (1 - \rho) = 1.508$, the standard error is 0.1139 pp at 32 months, and the smallest total length meeting the criterion is 26 months. For RQ3, Fisher's $z$ for a correlation of $r = 0.20$ gives $n = (2.802 / 0.2027)^2 + 3 = 194$ peak bank-months. For RQ4, a Cox model with 3 covariates and 10 events per variable needs 30 spike episodes.

**Table 3**
*Required and Available Sample Size by Research Question*

| RQ | Method | Key parameters | Required | Available | Verdict |
|---|---|---|---|---|---|
| RQ1 (level of ln*V*) | Power for a slope | $\beta = 0.60$; $\sigma_e = 1.0508$; $SD_x = 0.1258$ | 1,522 bank-months | 1,555 | Met (margin 33) |
| RQ1 (growth, $\Delta$ln*V*) | Power for a slope | Same, $SD_x = 0.0754$ | 4,239 bank-months | 1,489 | Not met |
| RQ2 | CI width (ITS) | Half-width 0.25 pp; $\sigma = 0.135$; $\rho = 0.203$ | 26 months | 32 | Met |
| RQ3 | Power for a correlation | $r = 0.20$ | 194 peak bank-months | 342 | Met |
| RQ4 | Events per variable | 3 covariates × 10 | 30 episodes | 69 | Met |

*Note.* Required N for the growth measure uses the same formula with the standard deviation of $\Delta$ln*V* in place of the within-bank SD of ln*V*.

The final sample for the panel is the largest of the requirements for the level measure, N = max(1,522; 26; 194; 30) = 1,522 bank-months, against 1,555 available. A consistency check changed one conclusion, however. RQ1 is estimated with month-over-month growth, whose standard deviation (0.075) is smaller than the within-bank SD of the volume level (0.126), and the same formula then requires about 4,239 bank-months, far above the 1,489 pairs available. The RQ1 growth estimates are therefore underpowered for an effect as small as $\beta = 0.60$, which is visible in the wide intervals reported later. It is stated as a limitation, and the final report will add September 2026 and beneficiary-side data and read RQ1 nulls as inconclusive. For RQ2, 32 months meet the 26-month requirement, but the smallest level change detectable with 80% power is 2.802 × 0.1139 = 0.319 pp, so RQ2 is reported as an estimate with an interval.

## Analysis

### Data Cleaning

Cleaning followed one rule: change a value only when the source clearly shows an error, flag judgment calls, and keep every real observation, including extreme ones, because spikes in technical declines are the phenomenon under study. All steps are in code/combine_remitter.py,

which rebuilds the panel from the raw files and reruns the integrity checks on every run. Table 4 lists each issue with its count, how it was detected, the treatment, and the reason.

**Table 4**
*Data Cleaning Log*

| Issue | Variables affected | Detection method | Treatment applied | Rationale |
|---|---|---|---|---|
| Missing values: 29 of 1,598 rows (1.8%) and 3 rows (0.2%); none in core fields | debit_reversal_mn; debit_reversal_success_pct | Null count by column | Left blank; the fields are not used in any model | Imputing a field that is unused adds assumptions without benefit; core variables are complete |
| Zero TD: 70 rows (4.4%) | td_pct | Value equals 0.00 | Kept; replaced by 0.0025 pp only when logits are computed | Reporting resolution is 0.01 pp, so a zero may be a rounded small value; logit needs a positive value |
| Outliers: 166 bank-months (10.7%) above the IQR fence of 2.18%; 104 flagged as spikes (6.7%) in 37 banks; maximum 36.31% | td_pct, volume_mn | IQR fence; robust z > 3 using bank median and MAD, with TD ≥ 1.0% | Retained and flagged (spike); sensitivity with winsorized TD planned | Extreme TD values are real failures, and deleting them would remove the signal; the spike flag makes them analyzable |
| Duplicates: 1 duplicate file (of 33); 2 duplicate listings of one bank in two months; 0 repeated bank-months after cleaning | All fields | Content fingerprint per file; duplicate (month, bank) check | Duplicate file dropped; duplicate listings combined by volume weighting | A duplicated month would inflate persistence and bias RQ4 |
| Inconsistent formats: 6 files (February–July 2025) store percentages as fractions, the rest as text | approved_pct, bd_pct, td_pct, debit_reversal_success_pct | Value range and text check | All converted to a 0–100 scale | Approved + BD + TD must equal 100 in every row (verified) |

| Issue | Variables affected | Detection method | Treatment applied | Rationale |
|---|---|---|---|---|
| Mislabeled month: 1 file whose sheet title differs from its file name (50 rows) | month | Compare title month with file month | Treated as August 2026 by file name and flagged; to be verified on the portal | Content differs from the July file, so it is not a duplicate |
| Name variants: 159 raw spellings | bank_raw, bank | Distinct-name count; rule-based regular expressions | Mapped to 63 names; one rename merged into an entity | One bank must be one unit in a bank panel |
| Non-bank members: 43 rows (2.7%) | is_bank | Name inspection (card product lines, payment firms) | Flagged is_bank = False; excluded from bank models | Different business model and failure profile |

*Note.* IQR = interquartile range; MAD = median absolute deviation. Counts come from code/interim_analysis.py.

After cleaning, the panel passes all checks: one file per month, no repeated bank within a month, and the percentage identity in every row. The workflow is reproducible: the script reads only the files in data/raw, writes only to data/processed, and the repository README lists the run order.

**EDA Results (Interpretation Emphasis)**

Table 5 summarizes the four main numeric variables for the 1,555 bank-months. TD is highly skewed: the median is 0.33% but the mean is 0.95% and the maximum is 36.31%, so a few observations carry much of the variation and a model on raw TD would be dominated by them. This is the first reason the study uses a fractional logit, which handles a skewed proportion near zero, and reports medians alongside means. Volume is skewed too (median 103 million, maximum 6,846 million), so volume enters models in logs.

**Table 5**
*Descriptive Statistics, Bank-Months (n = 1,555)*

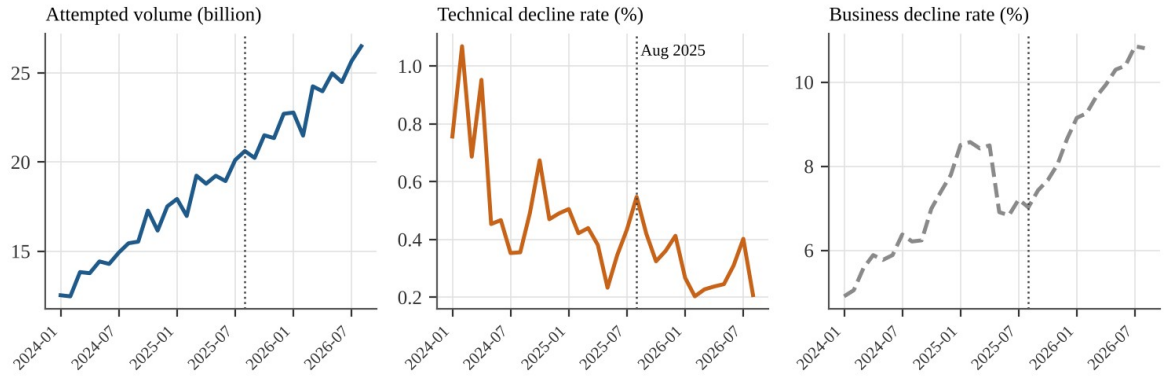
| Variable | Mean | SD | Minimum | Median | 99th pct. | Maximum |
|---|---|---|---|---|---|---|
| Attempted volume (million) | 392.1 | 776.5 | 14.1 | 103.4 | 4935.1 | 6845.8 |
| Approved share (%) | 90.51 | 4.39 | 54.14 | 91.80 | 96.38 | 97.18 |
| BD share (%) | 8.55 | 3.71 | 2.23 | 7.59 | 22.78 | 26.97 |
| TD share (%) | 0.95 | 1.91 | 0.00 | 0.33 | 10.11 | 36.31 |

*Note.* SD = standard deviation; pct. = percentile. The variation in TD splits into between-bank SD = 1.27 pp and within-bank SD = 1.51 pp.

Figure 4 shows the system-level picture. Attempted volume of the bank members roughly doubled over 32 months (12.5 to 26.5 billion, about 2.4% growth per month), while the volume-weighted TD rate fell from 0.76% in January 2024 to 0.21% in August 2026, with a high of 1.07% in 2024-02 and a low of 0.20% in 2026-02. The BD rate rose from 4.9% to 10.8%. The correlation of monthly volume with TD is −0.71 and with BD is 0.92, but both series trend over time, so these correlations mostly reflect shared trends and do not show that volume reduces TD. System-level growth did not coincide with worse technical reliability, so any RQ1 effect must be found in bank-level shocks.

**Figure 4**

*System-Level Trends in Volume, Technical Decline Rate, and Business Decline Rate*

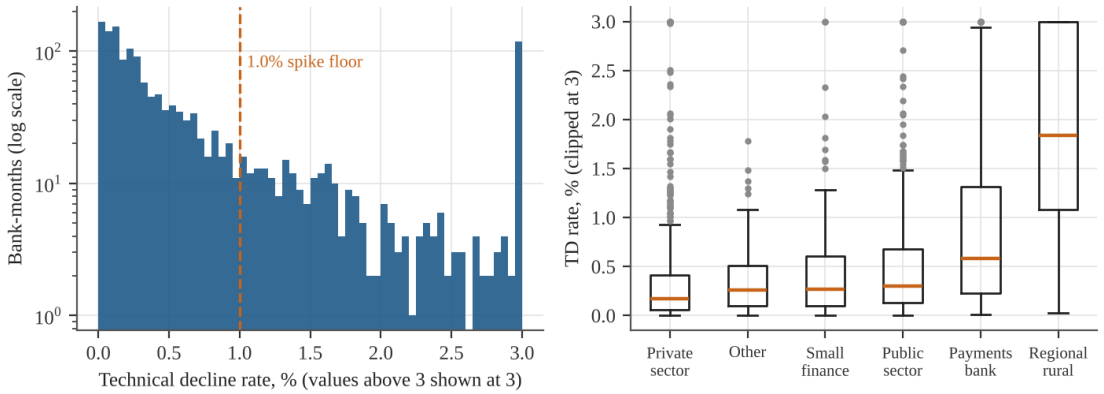



*Note.* Volume-weighted monthly values for bank members. The dotted line marks August 2025, when the NPCI API controls took effect. Source: author's calculation from NPCI (n.d.).

Figure 5 shows that the distribution of TD is heavily right-skewed and that bank type matters. Regional rural banks have a median bank-month TD of 1.84% and a mean of 2.88% (pooled rate 2.68%), against medians of 0.17% for private banks and 0.30% for public sector banks, although they handle only 1.6% of volume (Table 6). Payments banks sit between (median 0.58%, pooled 0.69%), and their spike rate of 11.3% of bank-months is the highest of all groups. Public sector banks carry 61% of volume with a moderate TD, so a system-level TD figure is mostly a statement about them. The pooled rate (total technical declines divided by total volume) is the better guide to what a user experiences: 0.43% for public sector banks, 0.18% for

private banks, and 0.41% for all banks together. A capacity rule targeting the system average would miss the small banks where TD is concentrated.

**Figure 5**
*Distribution of Technical Decline Rates and Differences by Bank Group*




*Note.* Left: histogram of bank-month TD (log count axis); values above 3% are shown at 3. Right: boxplots by bank group, ordered by median. Groups are assigned by rule from bank names. Source: author's calculation from NPCI (n.d.).

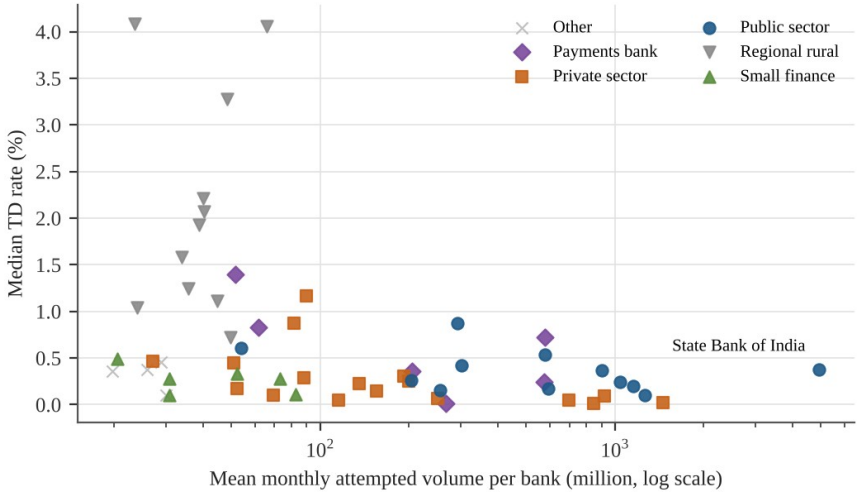
**Table 6**
*Technical Decline Rate and Spike Frequency by Bank Group*

| Group | Banks | Bank-months | Mean TD (%) | Median TD (%) | Pooled TD (%) | Spike rate (%) | Share of volume (%) |
|---|---|---|---|---|---|---|---|
| Public sector | 12 | 384 | 0.61 | 0.30 | 0.43 | 7.6 | 60.7 |
| Private sector | 17 | 537 | 0.43 | 0.17 | 0.18 | 3.9 | 28.4 |
| Small finance | 6 | 143 | 0.50 | 0.27 | 0.45 | 8.4 | 1.1 |
| Payments bank | 6 | 151 | 1.20 | 0.58 | 0.69 | 11.3 | 7.8 |
| Regional rural | 11 | 251 | 2.88 | 1.84 | 2.68 | 8.8 | 1.6 |
| Other | 4 | 89 | 0.36 | 0.26 | 0.36 | 3.4 | 0.4 |

*Note.* Mean and median TD are taken over bank-month rates (each row weighted equally); pooled TD is total technical declines divided by total attempted volume in the group. Spike = robust z-score of TD above 3 and TD at least 1.0%. "Other" combines foreign banks and a co-operative bank.

Figure 6 plots each bank's median TD against its average volume. The relationship is negative: larger banks have lower TD (Spearman correlation −0.51 across 56 banks). The ten largest banks handle 72% of volume and average a median TD of 0.16%, against 0.81% for the rest (averages of bank medians). A cross-bank regression of TD on volume would therefore be confounded by size, so the models use bank fixed effects.

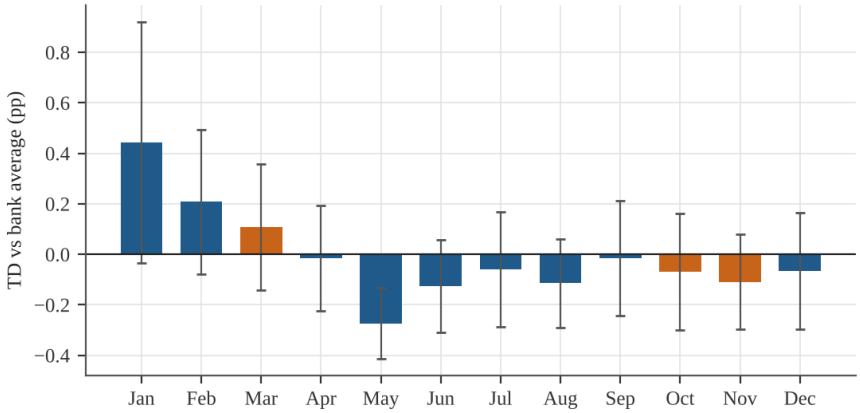
**Figure 6**
*Median Technical Decline Rate Against Average Volume, by Bank*




*Note.* Each point is one bank entity (n = 56); the horizontal axis is on a log scale. Source: author's calculation from NPCI (n.d.).

Figure 7 tests the calendar idea behind RQ3. After removing each bank's average, TD is highest in January (+0.44 pp) and February (+0.21 pp), slightly above average in March (+0.11 pp), and not elevated in the festive months (October −0.07 pp, November −0.11 pp); May is the lowest (−0.27 pp). Only May's interval excludes zero. January and February reflect early 2024, and 47 of the 104 spike-months fall in January–April 2024 (Figure 9). The data give little evidence of the hypothesized March and festive peaks.

**Figure 7**
*Technical Decline Rate by Calendar Month, Relative to Each Bank's Average*

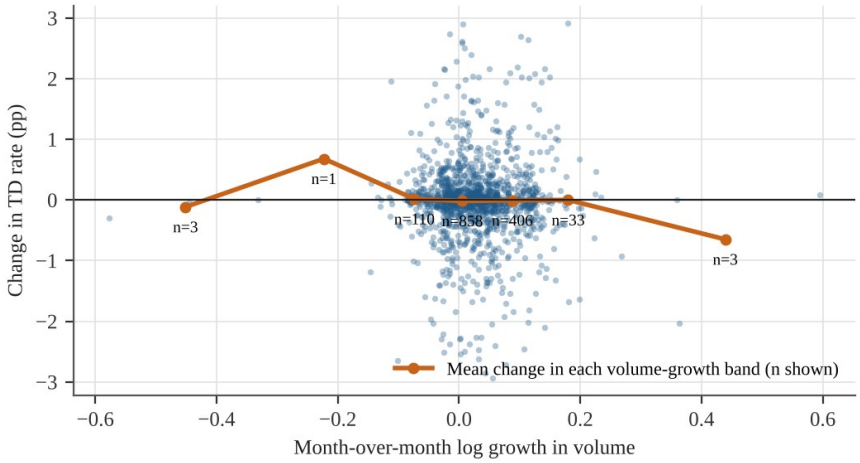



*Note.* Bars show the mean deviation from the bank average in pp with 95% intervals; orange bars are the hypothesized peak months (March, October, November). Source: author's calculation from NPCI (n.d.).

Figure 8 addresses RQ1 directly by relating monthly volume growth to the change in TD for the same bank. The cloud of points is centered on zero and the rank correlation is 0.003 ($p$ = 0.91, n = 1,489), so ordinary month-to-month growth, which has an SD of only 0.075, does not move TD in a visible way. The tails are different: in the 14 bank-months with volume growth above 0.20 (about 22%), TD rose by 1.06 pp on average, against −0.07 pp for the rest. That average is driven by three small banks (Baroda UP Gramin Bank and Rajasthan Marudhara Gramin Bank in March 2024, and Jio Payments Bank in March 2025), whose TD rose by 7.0, 4.1, and 6.6 pp. Three cases are anecdotal, but they suggest where an effect would appear: rare, large jumps at small banks around the March fiscal year-end, which an average-effect model has little power to find. NewPeak is also a weak indicator, since 56% of bank-months set a new maximum simply because volume trends upward.

**Figure 8**

*Monthly Volume Growth and the Change in Technical Decline Rate, Same Bank*

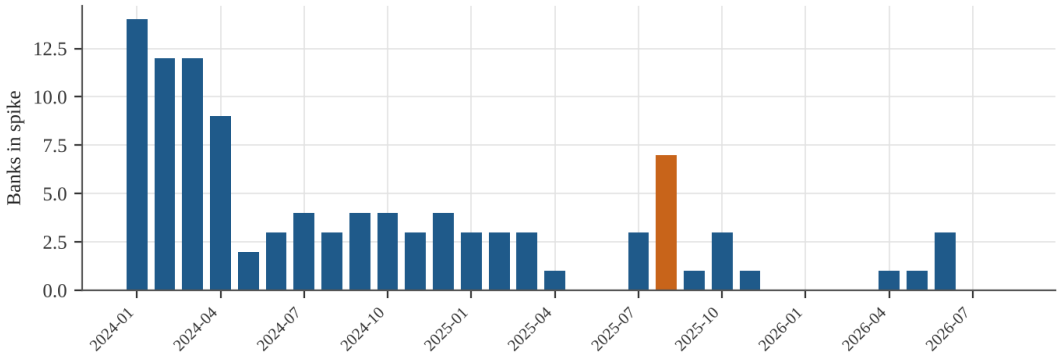



*Note.* Points are bank-months (|ΔTD| < 3 pp, |Δln*V*| < 0.8 shown); the line gives the mean change within volume-growth bands, with the number of observations. Source: author's calculation from NPCI (n.d.).

Figure 9 shows when spikes occur. They are concentrated at the start of the window (14, 12, 12, and 9 banks in spike in January–April 2024), thin out through 2025, and nearly disappear in early 2026. August 2025, the month the controls took effect, has a small cluster of 7 banks, but the 2024 cluster is larger and predates any control. Table 7 collects these insights and the decisions they imply. This pattern is the descriptive reason RQ2 is difficult: TD was already

falling when the controls arrived, so a before-and-after comparison confounds the intervention with a pre-existing downward trend.

**Figure 9**
*Number of Banks in a Technical Decline Spike, by Month*




*Note.* A bank is in a spike if its robust z-score of TD exceeds 3 and TD is at least 1.0%. The orange bar marks August 2025. Source: author's calculation from NPCI (n.d.).

**Table 7**
*EDA Insight Summary*

| Figure / table | What it shows | Key insight | Why it matters (link to RQ) | Decision / next step |
|---|---|---|---|---|
| Table 5 | Distribution of TD and volume | TD is skewed: median 0.33%, mean 0.95%, maximum 36.3% | Raw-scale linear models would be dominated by a few values (all RQs) | Fractional logit; report medians |
| Figure 4 | System trends | Volume rose 2.1× while TD fell from 0.76% to 0.21% | System-level growth did not worsen TD (RQ1); TD was already falling before August 2025 (RQ2) | Use bank-level changes; guard against trend |
| Figure 5, Table 6 | TD by bank group | Regional rural median 1.84% vs. 0.17% private; 1.6% of volume | TD is concentrated in small banks (RQ4, recommendations) | Bank fixed effects; group interactions |
| Figure 6 | Bank size and TD | Larger banks have lower TD (Spearman −0.51) | Cross-bank comparisons are confounded by size (RQ1) | Within-bank estimation |
| Figure 7 | Calendar months | No elevation in March or festive months; January–February reflect early 2024 | Little support for calendar peaks (RQ3) | Test peak with trend control; report null if so |
| Figure 8 | Growth vs. TD change | No average relation (ρ = 0.003); large jumps at three small banks coincide with TD rises | Effects, if any, are rare and bank-specific (RQ1) | Report with intervals; examine tail cases |

| Figure / table | What it shows | Key insight | Why it matters (link to RQ) | Decision / next step |
|---|---|---|---|---|
| Figure 9 | Spikes by month | 47 of 104 spike-months occur in Jan–Apr 2024 | Spikes are early-window; confounds before-after contrasts (RQ2, RQ4) | Placebo breaks; sensitivity without 2024 |

*Note.* pp = percentage points.

## Modelling

### Choice of Models with Justification

Models were chosen by problem type and by the research question each serves, with a baseline for every prediction task. RQ1–RQ3 are explanatory regression problems on a bounded proportion, RQ2 is a policy-timing problem on a single time series, RQ4 is a variance-and-duration problem, and the ML benchmark is a regression-and-classification problem for next-month TD.

**Model 1: Baselines (naive and bank-history mean).** The naive rule predicts that next month's TD equals this month's; the second baseline predicts the bank's average TD so far. A model is useful only if it beats them. Because TD is persistent ($\rho = 0.32$), the naive rule is demanding.

**Model 2: Fractional logit (quasi-binomial generalized linear model).** For RQ1 and RQ3, TD share is modeled with a logit link and bank fixed effects (plus month fixed effects for RQ1), with bank-clustered standard errors (SE) (Papke & Wooldridge, 1996). It suits a bounded proportion with many values near zero. With about 56 clusters, cluster-robust tests may over-reject, so a wild cluster bootstrap is planned (Cameron et al., 2008). The same family, with bank-group indicators and lags, is the predictive comparison model.

**Interrupted time series for RQ2.** Volume-weighted monthly TD is modeled with a level change at August 2025 under five specifications (step only; linear trend plus step; a 12-month window; and the log versions of the first two), with Newey–West standard errors (Newey & West, 1987) and a placebo-in-time test at all 21 candidate break months (Abadie et al., 2010; Bernal et al., 2017). BD serves as a placebo outcome.

**Variance, persistence, and survival models for RQ4.** A random-intercept model on logit TD gives the ICC, a pooled first-order autoregression gives persistence and a half-life, and Kaplan–Meier curves and a Cox model describe the duration of spike episodes (Cox, 1972; Kaplan & Meier, 1958).

**Model 3: XGBoost.** A gradient-boosted tree model (Chen & Guestrin, 2016) is used as the ML benchmark for next-month TD (regression) and next-month spikes (classification, with the positive class weighted). It can capture nonlinearities the logit misses but gives no effect estimates and may overfit 71 training spikes, so it is compared against simpler models.

## Features Included and Feature Engineering

Table 8 lists the features. Engineered features follow the research questions: log volume and its growth for RQ1, a new-peak indicator, calendar indicators for RQ3, and the bank's own TD history for persistence (RQ4). Bank group is coded by rule from names (public sector, private sector, small finance, payments, regional rural, other). All features for predicting month $t + 1$ use information available at month $t$ only. The historical mean is an expanding mean of the bank's own TD through month $t$, not a full-sample mean, which avoids leakage. Spike labels use full-sample bank medians, so they are an evaluation target and never an input feature. Models are always trained on months before the test month. Feature importance from XGBoost (last fit) ranks regional-rural group membership first (0.28), followed by current TD (0.14) and the bank's historical mean (0.09); volume growth ranks low (0.04), consistent with the EDA.

**Table 8**
*Feature Set and Usage*

| Feature | Original / engineered | Type | Reason for inclusion | Used in model(s) |
|---|---|---|---|---|
| $TD_t$, $TD_{t-1}$ | Original (lagged) | Numeric | Persistence of TD ($\rho = 0.32$) | Naive, logit, XGBoost |
| $BD_t$ | Original | Numeric | Business declines may signal load or user-side stress | Logit, XGBoost |
| $\ln V_t$ | Engineered (log) | Numeric | Bank size and exposure; removes skew | Logit, XGBoost |
| $\Delta \ln V_t$ | Engineered | Numeric | Sudden volume shift (RQ1) | All regression models |

| Feature | Original / engineered | Type | Reason for inclusion | Used in model(s) |
|---|---|---|---|---|
| $NewPeak_t$ | Engineered | Binary | Volume above previous maximum (RQ1) | All regression models |
| Bank history mean TD | Engineered (expanding) | Numeric | Bank-specific level; no leakage | Baseline, logit, XGBoost |
| sin, cos of target month; Peak | Engineered | Numeric / binary | Seasonality and calendar peaks (RQ3) | Logit, XGBoost |
| Bank group (5 indicators) | Engineered (rule) | Binary | TD level differs by group (Figure 5) | Logit, XGBoost |
| Post | Engineered | Binary | Intervention timing (RQ2, RQ3) | ITS, RQ3 logit |
| Bank, month fixed effects | Engineered | Dummies | Remove stable bank differences and common shocks | RQ1, RQ3 logit |

*Note.* Time-series (ITS) and survival models use the system series and the episode table, respectively.

## Evaluation Metrics (Include Formulae and Calculations)

Prediction is evaluated by rolling-origin validation: for each target month from January to August 2026, models are trained on all earlier months and predict that month (Bergmeir & Benítez, 2012), (code excerpts are in Appendix A), which mirrors how a bank or NPCI would use the model and keeps the test strictly after the training data. This gives 374 out-of-sample bank-months. Regression accuracy is measured by root mean squared error (RMSE) and mean absolute error (MAE) in pp. Spike detection is measured by precision, recall, and average precision (PR-AUC), because only 5 of the 374 test bank-months are spikes (1.3%), where accuracy would be misleading (Saito & Rehmsmeier, 2015). Table 9 gives the formulas with worked values for the naive baseline: its squared errors average 0.832, so RMSE = √0.832 = 0.912 pp, and it flags 5 spikes of which 2 are correct, so precision = 2/5 = 0.40 and recall = 2/5 = 0.40. For the explanatory models, effects are reported as average marginal effects (AME) with 95% confidence intervals, and ICC and half-life are computed from variance components.

**Table 9**
*Evaluation Metrics and Formulae*

| Metric | Formula | Interpretation | Used for |
|---|---|---|---|
| RMSE | $\sqrt{(1/n)\,\Sigma\,(y_i - \hat{y}_i)^2}$ | Typical error in pp, penalizing large misses | Regression benchmark |
| MAE | $(1/n)\,\Sigma\,\lvert y_i - \hat{y}_i \rvert$ | Average absolute miss in pp | Regression benchmark |

| Metric | Formula | Interpretation | Used for |
|---|---|---|---|
| Precision | TP / (TP + FP) | Share of flagged spikes that were real | Spike detection |
| Recall | TP / (TP + FN) | Share of real spikes that were flagged | Spike detection |
| PR-AUC | $AP = \Sigma_n (R_n - R_{n-1}) P_n$ | Ranking quality for rare spikes (base rate 0.013) | Spike detection |
| AME | mean[ $\hat{\beta} \cdot \hat{\mu}(1 - \hat{\mu})$ ] × 100 | Change in TD in pp per unit of the variable | RQ1, RQ3 |
| ICC | $\sigma^2_u / (\sigma^2_u + \sigma^2_e)$ | Share of logit-TD variance that is bank-specific | RQ4 |
| AR(1) half-life | $\ln(0.5) / \ln(\rho)$ | Months for a deviation to halve | RQ4 |
| Kaplan–Meier | $S(t) = \prod (1 - d_j / n_j)$ | Share of spikes still ongoing at month $t$ | RQ4 |

*Note.* TP, FP, FN = true positives, false positives, false negatives; $R$ = recall; $P$ = precision; $d_j$ = events and $n_j$ = spikes at risk at month $j$.

## Preliminary Results

### Preliminary Findings by Research Question

#### *RQ1: Volume shifts and TD*

Table 10 reports fractional logit models with bank and month effects on 1,489 bank-month pairs from 56 banks. Model A uses the two volume-shift measures only. A 10% volume jump is associated with a change of +0.029 pp in TD (95% CI −0.08 to +0.14), and a new peak with +0.054 pp (95% CI −0.15 to +0.26). Neither is statistically distinguishable from zero ($p$ = 0.59 and 0.60), so $H0_{1a}$ and $H0_{1b}$ are not rejected. Adding last month's TD (Model B) makes the point estimates larger (+0.093 pp for a 10% jump and +0.167 pp for a new peak; the latter has $p$ = 0.099) but the intervals still include zero, while the lag coefficient is large and significant (10.3, $p < .001$), confirming strong persistence. The result is a wide null, not evidence of no effect: the intervals are compatible with effects from slightly negative to a few tenths of a percentage point, and the power check in Table 3 shows that growth-based estimates are underpowered.

**Table 10**
*Preliminary Fractional Logit Estimates for RQ1 (Bank and Month Fixed Effects)*

| Model / variable | Coefficient (logit) | Cluster SE | *p* | AME (pp) | 95% CI for AME |
|---|---|---|---|---|---|
| A. $\Delta \ln V$ (per 10% jump) | 0.355 | 0.662 | 0.59 | +0.029 | [−0.08, 0.14] |
| A. NewPeak | 0.062 | 0.118 | 0.60 | +0.054 | [−0.15, 0.26] |
| B. $\Delta \ln V$ (per 10% jump) | 1.165 | 0.854 | 0.17 | +0.093 | [−0.04, 0.23] |
| B. NewPeak | 0.200 | 0.121 | 0.099 | +0.167 | [−0.03, 0.37] |
| B. $TD_{t-1}$ (share) | 10.26 | 1.49 | < .001 | — | — |

*Note.* Dependent variable: TD share. Model A n = 1,489, Model B n = 1,433. AME = average marginal effect in percentage points; intervals use cluster-robust standard errors and are not yet bootstrap-corrected.

### *RQ2: Timing of the API controls*

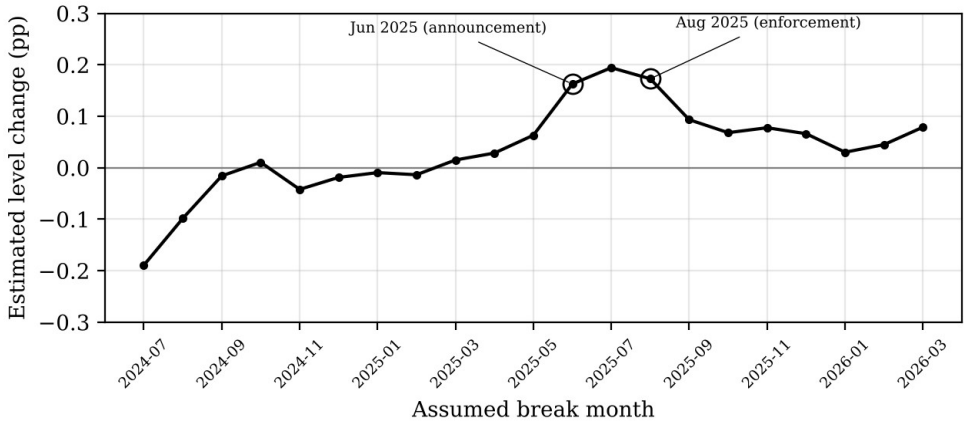
Table 11 and Figure 10 show that the August 2025 estimate depends on the specification. Without a trend (A and D), TD is lower after August 2025 (−0.205 pp, 95% CI −0.361 to −0.049), but the August step ranks only 20th of 21 break months, because earlier breaks give larger drops while TD was already falling. With a linear trend (B and E) the sign reverses (+0.173 pp), since TD sits above the extrapolated pre-trend, and the August estimate ranks 3rd, behind July 2025 (+0.194 pp) and July 2024 (−0.189 pp), with June 2025 (+0.163 pp) close behind. The 12-month window (C) shows a small decline of −0.116 pp. The pre-set decision rule requires agreement in sign across all five specifications and a top-two placebo rank; both fail, so the preliminary conclusion is that RQ2 is inconclusive on attribution. TD improved and flattened around mid-2025, but these data cannot separate the August 1 controls from that wider pattern. As a placebo, BD shows no level shift with a trend (−0.93 pp, 95% CI −2.70 to +0.84) but rises by 2.4 pp without a trend, because BD has risen steadily since 2024, so BD is a valid placebo only in the trend-adjusted form.

**Table 11**
*Specification Check for the August 2025 Level Change in TD*

| Specification | Estimate | 95% CI | Rank of August (of 21) |
|---|---|---|---|
| A. Step only (no trend) | −0.205 pp | [−0.361, −0.049] | 20 |
| B. Linear trend + step (planned) | +0.173 pp | [0.045, 0.301] | 3 |
| C. 12 months before vs. after, step | −0.116 pp | [−0.222, −0.011] | n/a |
| D. Log TD, step only | −37.7% | [−54.9%, −13.7%] | 19 |
| E. Log TD, linear trend + step | +30.2% | [2.8%, 64.8%] | 3 |

*Note.* Volume-weighted monthly TD, 32 months; Newey–West standard errors (lag 2) with *t* intervals. Rank is by absolute size among 21 candidate break months (1 = largest).

**Figure 10**
*Estimated Level Change in TD by Assumed Break Month (Linear Trend plus Step)*




*Note.* Each point re-fits the planned model with the break at that month. Circles mark June 2025 (announcement) and August 2025 (enforcement).

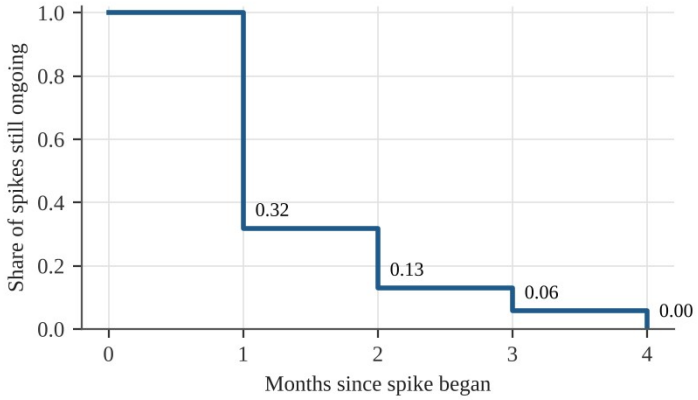
### *RQ3: Calendar peaks*

In a fractional logit with bank effects, a linear time trend, and the post-August-2025 indicator, the peak-month indicator has a coefficient of −0.025 (SE = 0.056, $p = 0.66$), an average marginal effect of −0.023 pp (95% CI −0.12 to +0.08) on 342 peak bank-months. The interaction of Peak with bank volume exposure is +0.040 (SE = 0.061, $p = 0.52$). Dropping March–April 2025 (near the April outage) leaves it at −0.063 (SE = 0.065). $H0_{3a}$ and $H0_{3b}$ are not rejected, consistent with Figure 7. Calendar peaks do not raise TD once the downward trend is controlled.

***RQ4: Persistence and heterogeneity***

The ICC of logit TD is 0.557 (between-bank variance 1.591, within-bank variance 1.264), so more than half of the variation in TD is attached to the bank, and $H0_{4a}$ is rejected. The within-bank autoregressive coefficient is 0.318, a half-life of 0.60 months, so deviations fade within about a month and $H0_{4b}$ is rejected. The rule identifies 104 spike-months in 37 banks, which form 69 episodes. All 69 recovered; the median duration is 1 month, the mean 1.51, and the longest 4 months, and 68% lasted a single month. Figure 11 gives the Kaplan–Meier curve: 0.32 of spikes were still ongoing after one month, 0.13 after two, and 0.06 after three. A Cox model with log volume, a peak-month start indicator, and public-sector status gives hazard ratios of 1.07 ($p$ = 0.68), 1.27 ($p$ = 0.41), and 0.63 ($p$ = 0.19) for recovery, so none is distinguishable from 1 and $H0_{4c}$ is not rejected on 69 episodes. Spikes are bank-specific and short.

**Figure 11**
*Kaplan–Meier Curve of Spike Duration*




*Note.* The vertical axis is the share of the 69 spike episodes still ongoing; labels show the survival estimate at each month. Source: author's calculation from NPCI (n.d.).

**Preliminary Model Performance**

Table 12 and Figure 12 compare the four approaches on the 374 out-of-sample bank-months from January to August 2026. Both the fractional logit (RMSE 0.708 pp) and XGBoost (0.721 pp) improve on the naive rule (0.912 pp) by about 22% in RMSE, while the bank-history mean does not (0.919 pp). On MAE, only XGBoost beats the naive rule (0.398 versus 0.420 pp), the logit is slightly worse (0.442 pp), and the history mean is clearly worse (0.582 pp). The three

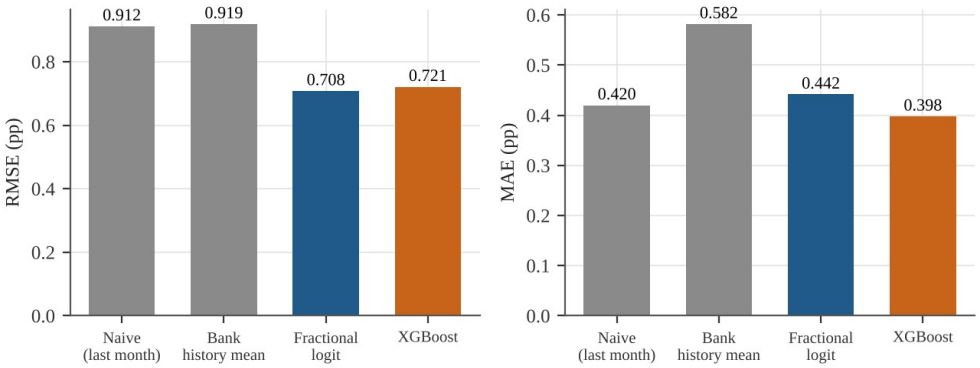
largest errors make up 33% of the naive baseline's squared error and 25% of XGBoost's; without them RMSE is 0.747 and 0.625, so the gain is concentrated in a few big movements.

**Table 12**
*Preliminary Model Performance Comparison (Rolling-Origin, January–August 2026)*

| Model | Feature set | Validation approach | RMSE (pp) | MAE (pp) | Key takeaway |
|---|---|---|---|---|---|
| Naive | $TD_t$ only | Rolling-origin, no fitting | 0.912 | 0.420 | Hard baseline; TD is persistent |
| Bank history mean | Expanding bank mean | Rolling-origin, no fitting | 0.919 | 0.582 | No gain over naive; level alone is not enough |
| Fractional logit | 15 features (Table 8) | Rolling-origin, refit monthly | 0.708 | 0.442 | Best RMSE; interpretable; MAE slightly worse than naive |
| XGBoost | 15 features (Table 8) | Rolling-origin, refit monthly | 0.721 | 0.398 | Best MAE; close to the logit on RMSE; untuned |

*Note.* n = 374 bank-months. Lower is better. XGBoost uses 300 trees, depth 3, learning rate 0.05, subsample 0.8, and seed 42 (no tuning yet).

**Figure 12**
*Out-of-Sample Error of the Baselines and Models, January–August 2026*




*Note.* Rolling-origin validation on 374 bank-months. Source: author's calculation from NPCI (n.d.).

Table 13 shows that spike detection is much weaker. With only 5 spikes in the test period, every number is imprecise. The naive rule, which flags banks that were in a spike last month, finds 2 of the 5 with 2 of 5 flags correct (precision and recall 0.40). The regression models flag every bank predicted at 1.0% or more, which catches 2 of 5 spikes but at a precision of 0.026 for the logit and 0.027 for XGBoost (76 and 74 flags), and the XGBoost classifier at a 0.5 threshold catches none of the 5 (17 flags). PR-AUC for every model is close to the base rate of 0.013 (naive 0.050, XGBoost regression score 0.039, XGBoost classifier 0.041). The model

comparison, then, supports one conclusion and rejects another: next-month TD levels can be predicted somewhat better than the naive rule, but next-month spikes cannot yet be predicted from public monthly data, and the naive "still in spike" rule is the best of a weak set.

**Table 13**
*Preliminary Spike Detection Performance (Rolling-Origin, January–August 2026)*

| Model (rule) | Flags raised | Precision | Recall | PR-AUC |
|---|---|---|---|---|
| Naive (in spike last month) | 5 | 0.40 | 0.40 | 0.050 |
| Fractional logit (predicted TD ≥ 1.0%) | 76 | 0.026 | 0.40 | 0.028 |
| XGBoost regression (predicted TD ≥ 1.0%) | 74 | 0.027 | 0.40 | 0.039 |
| XGBoost classifier (probability ≥ 0.5) | 17 | 0.00 | 0.00 | 0.041 |

*Note.* Test period has 5 spikes in 374 bank-months (base rate 0.013); the classifier was trained on 71 spikes before 2026.

### How the Results Answer the Research Questions

Taken together, the preliminary results answer the four questions as follows. For RQ1, average month-to-month volume shifts show no statistically detectable effect on bank-level TD, and the data are underpowered for effects as small as the planning value, so the question stays open; the only suggestive evidence is a handful of large March jumps at small banks. For RQ2, TD fell and flattened around mid-2025, but the August 2025 timing is not distinguishable from nearby breaks, so the effect of the API controls cannot be attributed from these data; finer data are needed. For RQ3, March and festive months do not raise TD once the trend is controlled. For RQ4, TD is strongly bank-specific (ICC 0.56) and spikes are short (median one month), so decisions should target individual banks, particularly small regional and payments banks, and use short monitoring windows. These results are preliminary.

### Interim Limitations and Risks

- **Low power for RQ1 growth effects:** required N is about 4,239 bank-months against 1,489 available. Mitigation: add September 2026 and beneficiary-side data, report intervals, treat nulls as inconclusive, and use the level-based design as primary.

- **RQ2 depends on specification, and one date and one system allow no untreated group:** mitigated by five specifications, placebo breaks, a pre-set decision rule, and a dose-response and difference-in-differences extension if data allow.
- **Few spikes in the test period (5):** spike-detection metrics are noisy. Mitigation: report them with rolling-origin results, add months, and avoid strong claims.
- **Data issues:** a judgment-based bank rename (0.8% of bank-months), a mislabeled August 2026 file to verify, 70 zero values that may be rounding, a changing top-50 list, and NPCI's own TD definition. Mitigation: sensitivity checks (Table 4 and the entity crosswalk), verification against the portal, and an explicit unbalanced-panel discussion.
- **Model limitations:** cluster-robust tests with about 56 banks may over-reject (wild bootstrap planned); the logit imposes a functional form; XGBoost is untuned and may overfit 71 training spikes; bank groups are coded by rule, not from an official register.
- **Validity:** results describe associations between banks and months, not individual users, and the observational design cannot prove cause, so recommendations are guidance for further analysis with transaction-level data.

**Next Steps (for Final Report)**

- Complete robustness checks: leave-one-bank-out, alternative spike thresholds ($z > 2.5$ with TD $\geq 0.75\%$), winsorized TD, and alternative floors for zero TD.
- Add wild cluster bootstrap p-values for RQ1 and RQ3, and a bank-group interaction for RQ1.
- Add September 2026 and beneficiary-side tables; use them for a difference-in-differences contrast for RQ2 and for more power in RQ1.
- Tune XGBoost with rolling-origin validation, compare with the logit again, and report calibration for the spike classifier.
- Verify the bank rename and the August 2026 file label against official sources.
- Write stakeholder recommendations for NPCI, banks, and oversight teams, conditional on the final estimates, and complete the final report and repository README.

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# Appendix A

## Code Excerpts

The complete code is in the repository (code/interim_analysis.py). Excerpts used to produce Table 10 and Table 12 are shown below. The analysis was run in Python with pandas, statsmodels, scikit-learn, and XGBoost, with random seed 42.

```
# Rolling-origin benchmark (code/interim_analysis.py)
for m in test_months:                       # 2026-01 ... 2026-08
    tr = M[M.next_month < m]; te = M[M.next_month == m]   # train strictly before
the test month
    preds["naive"].append(te.td_pct.values)
    g1 = sm.GLM((tr.next_td/100).clip(0, .5), Xtr, family=Binomial()).fit()
    xr = xgb.XGBRegressor(n_estimators=300, max_depth=3, learning_rate=0.05,
                          subsample=0.8, colsample_bytree=0.8, random_state=42)
    xr.fit(tr[FEATS], tr.next_td.clip(upper=10))

# RQ1 fractional logit with entity and month fixed effects, clustered SE
X = pd.get_dummies(df[["entity","month"]], drop_first=True)
m = sm.GLM(df.p, sm.add_constant(pd.concat([df[["dlnV","newpeak"]], X], axis=1)),
           family=sm.families.Binomial()).fit(cov_type="cluster",
cov_kwds={"groups": codes})
```